Unit: Unit 0: Foundations of Algebra
Topic: GCF LCM

Name: ______________
Date: __________

Advanced (Carolina Reaper)

Q1. Find the GCF of 12 and 15 using factors.
(A) 1
(B) 3
(C) 5
(D) 6
(E) 12

Q2. A bookshelf has 18 shelves, and each shelf can hold 12 books. What is the LCM of 18 and 12?
(A) 36
(B) 48
(C) 54
(D) 60
(E) 72

Q3. Find the LCM of 8 and 10 using prime factorization.
(A) 20
(B) 30
(C) 40
(D) 50
(E) 60

Q4. A music store sells guitar strings in packs of 6 or 8. What is the GCF of 6 and 8?
(A) 1
(B) 2
(C) 4
(D) 6
(E) 8

Q5. A group of friends want to go on a road trip that is 240 miles long. If they drive 15 miles per hour, how many hours will it take them to complete the trip, assuming they drive for 8 hours a day? Use GCF and LCM to solve.
(A) 12 hours
(B) 16 hours
(C) 18 hours
(D) 20 hours
(E) 24 hours

Q6. Find the GCF of 24 and 30 using prime factorization.
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

Q7. A bakery sells bread in loaves of 12 or 15. What is the LCM of 12 and 15?
(A) 36
(B) 48
(C) 60
(D) 72
(E) 120

Q8. A water tank can hold 120 gallons of water. If 8 gallons of water are added to the tank every hour, how many hours will it take to fill the tank? Use GCF and LCM to solve.
(A) 10 hours
(B) 12 hours
(C) 15 hours
(D) 18 hours
(E) 20 hours

Open Ended Questions (FRQ)

Q9. A group of friends want to go on a camping trip that is 180 miles long. If they drive 12 miles per hour, how many hours will it take them to complete the trip, assuming they drive for 8 hours a day? Use GCF and LCM to solve. Show all work and explain your answer.

Q10. A store sells T-shirts in packs of 9 or 12. If a customer wants to buy 36 T-shirts, how many packs of each size should they buy to minimize the total number of packs? Use GCF and LCM to solve. Show all work and explain your answer.

Chef's Tips

- Tip 1: Emphasize the importance of prime factorization in finding GCF and LCM.
- Tip 2: Encourage students to use real-world examples to illustrate the concepts of GCF and LCM.
- Tip 3: Remind students to show all work and explain their answers clearly.